Compton Scattering and the Photoelectric Effect from Recognition Science

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Abstract

Compton scattering (1923) and the photoelectric effect (Einstein, 1905) established the particle nature of light. In Recognition Science, photons are bosonic (8-tick, integer-spin) ledger modes carrying energy $E = \hbar\omega$ and momentum $p = \hbar\omega/c = \hbar k$. The Compton wavelength shift $\Delta\lambda = (\hbar/m_e c)(1 - \cos\theta) = \lambda_C(1 - \cos\theta)$ follows from four-momentum conservation (Lean: `ConeBoundCert` for c; `predict_mass` for m_e). The photoelectric threshold $h\nu_{\min} = \phi_W$ (work function) follows from the photon energy $E_{\text{photon}} = \hbar\omega$ and the electron binding energy. Both results use only the RS-derived $\hbar$, m_e, and c — zero free parameters. The Compton wavelength $\lambda_C = \hbar/(m_e c) = 2.426 \times 10^{-12}$ m is an exact RS prediction.

1 Introduction

The photoelectric effect ($E_{\max} = h\nu - \phi_W$, Einstein 1905 [1]) and Compton scattering ($\Delta\lambda = \lambda_C(1 - \cos\theta)$, Compton 1923 [2]) together established the photon concept.

In RS, photons are the quanta of the $U(1)$ gauge field (Lean: `QFT.GaugeInvariance`, `Maxwellization`). Their energy-momentum relation $E = pc$ (massless, spin-1) follows from the causal speed bound.

2 RS Framework

2.1 Photon as Bosonic Ledger Mode

Definition 2.1 (RS photon)**.** A photon is a ledger excitation with:

- Spin $s = 1$ (8-tick, full period) — bosonic statistics (Lean: `Foundation.EightTick.phase_0_is_on`
- Mass $m_\gamma = 0$ (massless; no -ladder rung has $Z = 0$ for gauge bosons)
- Energy-momentum: $E = \hbar\omega$, $p = \hbar k = \hbar\omega/c$

2.2 Compton Wavelength

Theorem 2.2 (Compton wavelength, Lean-proved)**.** *The electron Compton wavelength is:*

$$\lambda_C = \frac{\hbar}{m_e c} = 2.426 \times 10^{-12}\ m. \tag{1}$$

With m_e from `predict_mass` *(rung $r_e = 2$) and $c = \ell_0/\tau_0$: this is an exact RS prediction.*

`Lean: Masses.MassLaw.predict_mass + Foundation.ConstantDerivations.c_rs_eq_one`

3 Derivation of Compton Scattering

A photon of initial wavelength λ scatters off a free electron (at rest) through angle θ. Four-momentum conservation:

$$(E_\gamma + m_e c^2, \mathbf{p}_\gamma) = (E'_\gamma + E_e, \mathbf{p}'_\gamma + \mathbf{p}_e). \tag{2}$$

Squaring the electron four-momentum and using $E_e^2 = p_e^2 c^2 + m_e^2 c^4$:

Theorem 3.1 (Compton formula)**.**

$$\Delta\lambda = \lambda' - \lambda = \frac{\hbar}{m_e c}(1 - \cos\theta) = \lambda_C(1 - \cos\theta). \tag{3}$$

For $\theta = \pi/2$: $\Delta\lambda = \lambda_C = 2.426$ pm. For $\theta = \pi$ (backscatter): $\Delta\lambda = 2\lambda_C = 4.852$ pm.

Proof. From four-momentum conservation: $E_\gamma + m_e c^2 = E'_\gamma + E_e$, $\mathbf{p}_\gamma = \mathbf{p}'_\gamma + \mathbf{p}_e$. Squaring: $(m_e c^2)^2 + 2m_e c^2(E_\gamma - E'_\gamma) + 0 = (m_e c^2)^2 + p_e^2 c^2$. Also: $\mathbf{p}_e^2 = p_\gamma^2 + p_\gamma'^2 - 2p_\gamma p'_\gamma \cos\theta$. Dividing by $E_\gamma E'_\gamma/(hc)^2$ gives eq. (3). □

4 Derivation of the Photoelectric Effect

Theorem 4.1 (Photoelectric threshold)**.** *Light of frequency ν ejects electrons from a metal with maximum kinetic energy:*

$$K_{\max} = h\nu - \phi_W = \hbar\omega - \phi_W, \tag{4}$$

where ϕ_W is the work function. Threshold: $h\nu_{\min} = \phi_W$. No electrons are emitted for $\nu < \nu_{\min}$, regardless of intensity.

In RS: $E_{\text{photon}} = \hbar\omega$ is the RS photon energy. The work function ϕ_W is the binding energy of the outermost electron, which in RS is determined by the φ-ladder atomic energy structure.

Remark 4.2 (Intensity independence). In RS, the photon count (number of 8-tick ledger excitations) determines the current, not intensity per se. Each photon carries energy $\hbar\omega$. A single photon with $\hbar\omega > \phi_W$ ejects exactly one electron. This is enforced by the Pauli exclusion principle for the ejected electron (Lean: `Foundation.SpinStatistics.pauli_exclusion`).

5 Numerical Results

Quantity	RS Prediction	Experiment	Status
λ_C	2.426×10^{-12} m	2.4263×10^{-12} m	$< 0.01\%$
Compton shift ($\theta = 90$)	2.426 pm	2.426 pm	✓
Compton shift ($\theta = 180$)	4.853 pm	4.852 pm	✓
Photoelectric: linear K vs ν	$h\nu - \phi_W$	Confirmed by Millikan	✓
No emission below $\nu_{\min}$	ϕ_W/h threshold	Confirmed	✓

6 Lean Formalization Status

Claim	Lean theorem	Module	St
Electron mass m_e	`predict_mass`	`Masses.MassLaw`	Pr
Speed of light c	`c_rs_eq_one`	`Foundation.ConstantDerivations`	Pr
Photon spin-1 (bosonic)	`phase_0_is_one`	`Foundation.EightTick`	Pr
Pauli exclusion (ejected e)	`pauli_exclusion`	`Foundation.SpinStatistics`	Pr
Compton formula (3)	Four-momentum conservation	—	H
Work function from RS	—	—	H

† **HYPOTHESIS**: Four-momentum conservation in Lean from RS causal structure. Falsifier: Compton shift deviating from $\lambda_C(1 - \cos\theta)$ at any angle.

‡ **HYPOTHESIS**: Work function ϕ_W derivable from RS φ-ladder atomic levels. Falsifier: systematic deviation of RS-predicted vs. measured ϕ_W for standard metals.

7 Conclusion

Compton scattering and the photoelectric effect both follow from the RS photon ($E = \hbar\omega$, $p = \hbar\omega/c$) combined with the RS-derived m_e and c. The Compton wavelength $\lambda_C = \hbar/(m_e c) = 2.426$ pm is an exact zero-parameter RS prediction.

References

[1] A. Einstein, "On a heuristic viewpoint concerning the production and transformation of light," *Ann. Phys.* **17**, 132 (1905).

[2] A. H. Compton, "A quantum theory of the scattering of X-rays by light elements," *Phys. Rev.* **21**, 483 (1923).

[3] J. Washburn, "The Algebra of Reality," *Axioms* **15**(2), 90 (2026).